

# Pairwise maximal monotonicity does not force maximal multi-marginal $c$ -monotonicity

Candidate counterexample packet for arXiv:1901.03777

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**Status.** Candidate counterexample, likely valid, awaiting human review. The construction answers the source question as stated; it does not address a strengthened version requiring the global set to be closed.

## 1 Source question

Bartz, Bauschke, Phan, and Wang study the cost

$$c(x_1, \dots, x_N) = \sum_{1 \leq i < j \leq N} \langle x_i, x_j \rangle$$

on a product of real Hilbert spaces. If  $\Gamma \subset H^N$ , write  $\Gamma_{i,j}$  for its projection onto coordinates  $i, j$ , viewed as the graph of a relation from  $H$  to  $H$ . After Proposition 5.9, they ask whether maximal monotonicity of every  $\Gamma_{i,j}$  implies maximal  $c$ -monotonicity of  $\Gamma$  [1].

**Transcription.** “To the best of our knowledge, the question whether the maximal monotonicity of the  $\Gamma_{i,j}$ ’s implies the maximal  $c$ -monotonicity of  $\Gamma$  is still open.”

## 2 Definitions used

A relation  $M \subset H \times H$  is monotone when

$$\langle x - y, u - v \rangle \geq 0 \quad ((x, u), (y, v) \in M),$$

and maximally monotone when it has no proper monotone extension. Minty’s range characterization says that a monotone relation  $M$  is maximally monotone exactly when

$$\{x + u : (x, u) \in M\} = H.$$

We use this in precisely the form recalled in [1, Fact 1.3].

For the above cost,  $\Gamma$  is  $c$ -monotone if and only if, for every nonempty proper  $K \subset \{1, \dots, N\}$  and every  $x, y \in \Gamma$ ,

$$\left\langle \sum_{i \in K} (x_i - y_i), \sum_{j \notin K} (x_j - y_j) \right\rangle \geq 0.$$

This is [1, Lemma 2.3]. A  $c$ -monotone set is maximally  $c$ -monotone if it has no proper  $c$ -monotone extension.

$\Gamma_{i,j}$  are monotone. With the exception of Example 5.4, in all of our examples of  $c$ -monotone sets in this section the set  $\Gamma$  had monotone two-marginal projections  $\Gamma_{i,j}$ . Thus, a natural question is: *How does the monotonicity and maximal monotonicity of the two-marginal projections  $\Gamma_{i,j}$  relate to the  $c$ -monotonicity and maximal  $c$ -monotonicity of  $\Gamma$ ?*

**Proposition 5.9** *Let  $X_i = \mathbb{R}^d$  for each  $1 \leq i \leq N$ . Let  $\Gamma \subseteq X$  be a set. Suppose that for each  $1 \leq i < j \leq N$  the set  $\Gamma_{i,j}$  is monotone. Then  $\Gamma$  is  $c$ -monotone.*

*Proof.* The mapping  $A_K$  is monotone if and only if for every  $(x_1, \dots, x_N), (y_1, \dots, y_N) \in \Gamma$ ,

$$0 \leq \left\langle \sum_{i \in K} x_i - y_i, \sum_{j \notin K} x_j - y_j \right\rangle.$$

Since the right-hand side is equal to  $\sum_{j \notin K} \langle x_i - y_i, x_j - y_j \rangle$  and since, by the monotonicity of  $\Gamma_{i,j}$ ,  $0 \leq \langle x_i - y_i, x_j - y_j \rangle$ , we see that  $A_K$  is monotone. ■

To the best of our knowledge, the question whether the maximal monotonicity of the  $\Gamma_{i,j}$ 's implies the maximal  $c$ -monotonicity of  $\Gamma$  is still open.

Finally, we note that the maximal  $c$ -monotonicity of  $\Gamma$  does not imply the maximal monotonicity of the  $\Gamma_{i,j}$ 's even when the  $\Gamma_{i,j}$ 's are monotone. Indeed, in Example 5.5, we see that although  $\Gamma$  is maximally  $c$ -monotone,  $\Gamma_{i,j}$  is a singleton for all  $3 \leq i < j \leq N$ , thus  $\Gamma_{i,j}$  is monotone but not maximally monotone. Even in the case  $N = 3$ ,  $\Gamma_{1,3}$  is a proper subset of the graph of the zero mapping whenever  $\Gamma$  is generated by a maximally monotone mapping  $A$  without a full domain. We conclude in this case that  $\Gamma$  is maximally  $c$ -monotone, however,  $\Gamma_{1,3}$  is not maximally monotone.

Figure 1: The exact question and preceding proposition, source PDF page 22.

### 3 Idea of the proof

The construction starts with three coordinatewise monotone maps  $T_1, T_2, T_3: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  which partition the identity. There is one coordinate for each pair 12, 13, 23. On the coordinate dedicated to pair  $ij$ , both  $T_i$  and  $T_j$  use a soft-threshold map and are therefore constant on an interval around zero. The remaining map carries the missing coordinate, keeping  $T_1 + T_2 + T_3 = \text{Id}$ .

Every pair sum is still onto, so Minty's theorem makes every pair projection maximally monotone. The flat interval supplies a different nonzero parameter which reproduces the deleted zero pair for each of the three projections. Thus removing the global origin creates a genuine  $c$ -monotone hole which none of the pair projections sees.

### 4 Counterexample

**Theorem 1.** *For  $N = 3$  and  $H = \mathbb{R}^3$ , there exists  $\Gamma \subset H^3$  such that every two-marginal projection  $\Gamma_{i,j}$  is maximally monotone, while  $\Gamma$  is not maximally  $c$ -monotone.*

*Proof.* Define the soft-threshold map

$$\rho(t) = \text{sgn}(t)(|t| - 1)_+ = \begin{cases} t + 1, & t < -1, \\ 0, & -1 \leq t \leq 1, \\ t - 1, & t > 1, \end{cases}$$

and set

$$\alpha(t) = \frac{1}{3}\rho(t), \quad \beta(t) = t - \frac{2}{3}\rho(t).$$

Both  $\alpha$  and  $\beta$  are continuous and nondecreasing: away from the two breakpoints their slopes are respectively  $1/3, 0, 1/3$  and  $1/3, 1, 1/3$ . Moreover,

$$2\alpha(t) + \beta(t) = t. \quad (1)$$

Use coordinates  $s = (s_{12}, s_{13}, s_{23})$  on  $H = \mathbb{R}^3$ , and define

$$\begin{aligned} T_1(s) &= (\alpha(s_{12}), \alpha(s_{13}), \beta(s_{23})), \\ T_2(s) &= (\alpha(s_{12}), \beta(s_{13}), \alpha(s_{23})), \\ T_3(s) &= (\beta(s_{12}), \alpha(s_{13}), \alpha(s_{23})). \end{aligned}$$

Equation (1) gives

$$T_1 + T_2 + T_3 = \text{Id}_H. \quad (2)$$

Let

$$\tilde{\Gamma} = \{(T_1(s), T_2(s), T_3(s)) : s \in H\}, \quad \Gamma = \tilde{\Gamma} \setminus \{(0_H, 0_H, 0_H)\}.$$

The deleted point has the unique parameter  $s = 0$ , by (2).

We first check pairwise monotonicity. Fix  $s, t \in H$ . On each scalar coordinate, every  $T_i$  uses either  $\alpha$  or  $\beta$ . Since both maps are nondecreasing, the product of their increments is nonnegative. Adding the three coordinate products yields, for  $i \neq j$ ,

$$\langle T_i(s) - T_i(t), T_j(s) - T_j(t) \rangle \geq 0. \quad (3)$$

Therefore every pair projection  $\tilde{\Gamma}_{i,j}$  is monotone. For each cut in the three-marginal definition, the required inner product is a sum of two expressions in (3); hence  $\tilde{\Gamma}$  is  $c$ -monotone. Its subset  $\Gamma$  is also  $c$ -monotone.

We next check pairwise maximality. Put

$$\phi = 2\alpha = \frac{2}{3}\rho, \quad \psi = \alpha + \beta = \text{Id}_{\mathbb{R}} - \frac{1}{3}\rho.$$

The map  $\phi$  is onto because  $\rho$  is onto. The map  $\psi$  has the explicit form

$$\psi(t) = \begin{cases} \frac{2}{3}t - \frac{1}{3}, & t < -1, \\ t, & -1 \leq t \leq 1, \\ \frac{2}{3}t + \frac{1}{3}, & t > 1, \end{cases}$$

so it is also onto. Directly from the definitions,

$$\begin{aligned} (T_1 + T_2)(s) &= (\phi(s_{12}), \psi(s_{13}), \psi(s_{23})), \\ (T_1 + T_3)(s) &= (\psi(s_{12}), \phi(s_{13}), \psi(s_{23})), \\ (T_2 + T_3)(s) &= (\psi(s_{12}), \psi(s_{13}), \phi(s_{23})). \end{aligned}$$

Each map is therefore onto  $H$ . Equivalently, for every  $i < j$ ,

$$\{x_i + x_j : (x_i, x_j) \in \tilde{\Gamma}_{i,j}\} = H.$$

Together with monotonicity, Minty's theorem implies that every  $\tilde{\Gamma}_{i,j}$  is maximally monotone.

It remains to see that deletion of the global origin changes none of these pair projections. Every pair value having a nonzero parameter remains in  $\Gamma_{i,j}$ . The pair value generated at the deleted

parameter  $s = 0$  is  $(0_H, 0_H)$ . Because  $\rho$  vanishes on  $[-1, 1]$ , this same pair value is generated by the following nonzero parameters:

| pair     | replacement parameter |
|----------|-----------------------|
| $(1, 2)$ | $(1/2, 0, 0)$         |
| $(1, 3)$ | $(0, 1/2, 0)$         |
| $(2, 3)$ | $(0, 0, 1/2)$ .       |

For instance, at  $s = (1/2, 0, 0)$  one has  $T_1(s) = T_2(s) = 0_H$ , while  $T_3(s) = (1/2, 0, 0)$ , so this is not the deleted triple. The other two checks are identical. Consequently,

$$\Gamma_{i,j} = \tilde{\Gamma}_{i,j} \quad (1 \leq i < j \leq 3),$$

and all three projections of  $\Gamma$  are maximally monotone.

Finally,  $\tilde{\Gamma}$  is a proper  $c$ -monotone extension of  $\Gamma$ . Thus  $\Gamma$  is not maximally  $c$ -monotone, completing the counterexample.  $\square$

## 5 Verification notes

The proof uses no numerical assumption. The included script `code/verify_counterexample.py` is a regression check for the piecewise identities. It tests all breakpoints, explicit inverses for the three pair-sum maps, the three deletion witnesses, and 50,000 seeded random pairs. It reported no contradiction.

The most important human checks are:

1. equation (3), where coordinatewise monotonicity is converted into Hilbert-space pair monotonicity;
2. the use of Minty’s range criterion after proving pair-sum surjectivity;
3. the fact that a different nonzero replacement parameter is allowed for each pair projection.

## 6 Novelty check and limitations

A bounded search on July 21, 2026 covered the exact quoted question, the arXiv id, the paper title, close combinations of “two-marginal projections”, “maximal monotonicity”, and “maximal  $c$ -monotonicity”, local run indexes and full text, and citation-facing results. It found no explicit resolution of this exact pair-projection question. This is evidence only for “no answer found in the bounded search”, not a definitive priority claim.

The constructed  $\Gamma$  is not closed: the mechanism is a single deleted global point invisible to every pair projection. The source question places no closedness condition on  $\Gamma$ , so the example answers it as written. Whether the implication holds under an additional closedness or pairwise-join completeness assumption remains outside this result.

## 7 Human review recommendation

Send to a monotone-operator or multi-marginal optimal-transport expert. The formal verification is short and self-contained. If accepted, the natural next question is whether a closed counterexample exists or whether closedness restores the implication.

## References

- [1] S. Bartz, H. H. Bauschke, H. M. Phan, and X. Wang, *Multi-marginal maximal monotonicity and convex analysis*, Mathematical Programming **185** (2021), 385–408; [arXiv:1901.03777v2](#).